

Guidelines in HCI

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Norman's 7 principles of human-computer interaction (HCI)

- To assess the interaction between human and computers, Donald Norman in 1988 proposed seven principles. He proposed the seven stages that can be used to transform difficult tasks.
- Norman's 7 principles of human-computer interaction (HCI) are a set of guidelines that aim to improve the usability and user experience of computer systems. These principles are as follows:
 1. Use both knowledge in world & knowledge in the head.
 2. Simplify task structures.
 3. Make things visible.
 4. Get the mapping right (User mental model = Conceptual model = Designed model).
 5. Convert constraints into advantages (Physical constraints, Cultural constraints, Technological constraints).
 6. Design for Error.
 7. When all else fails – Standardize.

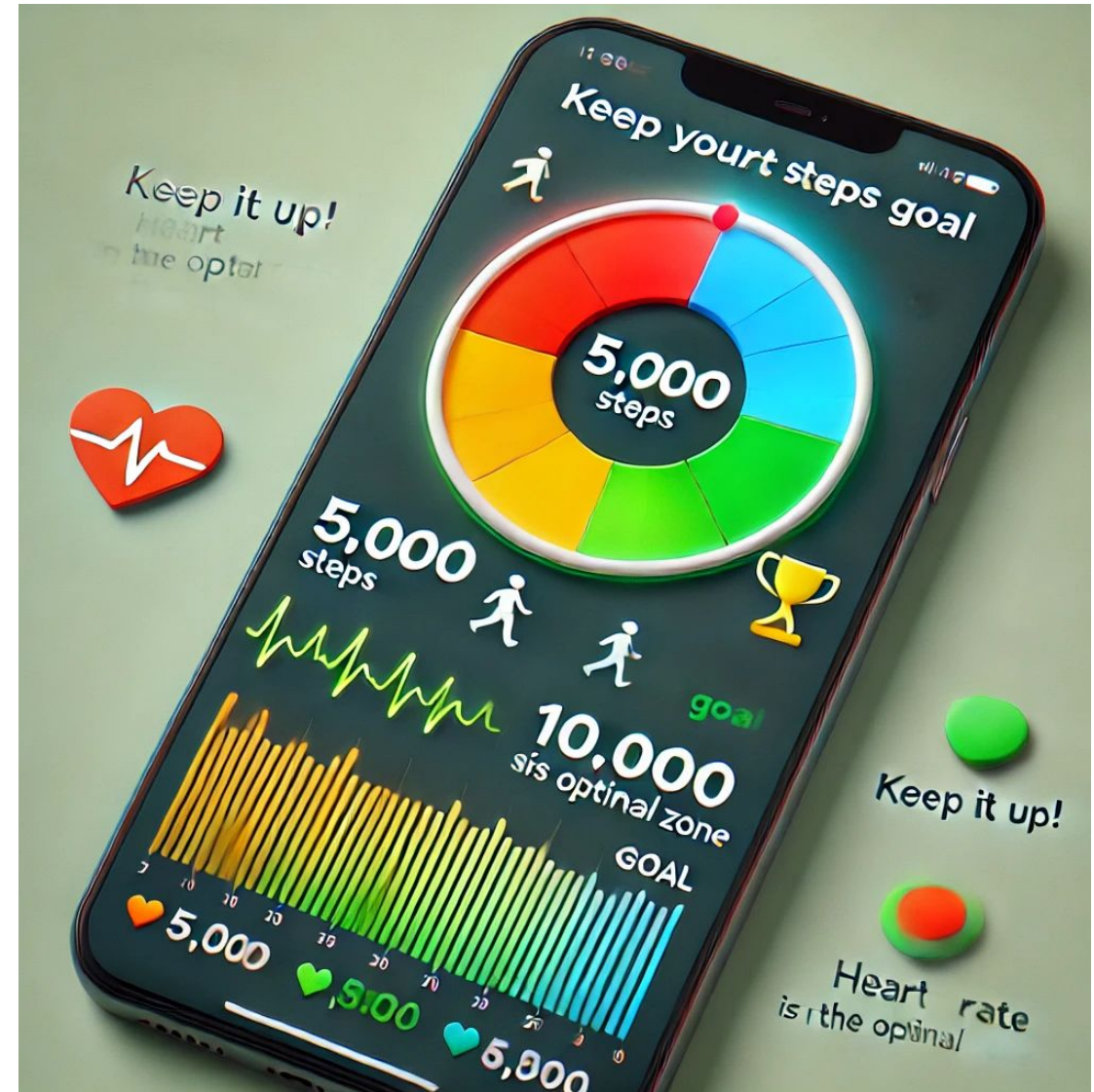
Use both knowledge in world & knowledge in the head.

Knowledge in the Head: Refers to information and skills a user already knows or has internalized, such as habits, memories, or learned behaviors.

Example: Sarah knows from past experience that 10,000 steps is a good daily goal for her health, so she remembers this goal without needing constant reminders.

Knowledge in the World: Refers to cues, symbols, or information present in the environment (interface) that help guide user actions.

Example: The app displays a progress bar showing how close Sarah is to her step goal, making it easy for her to check her progress without relying on memory.

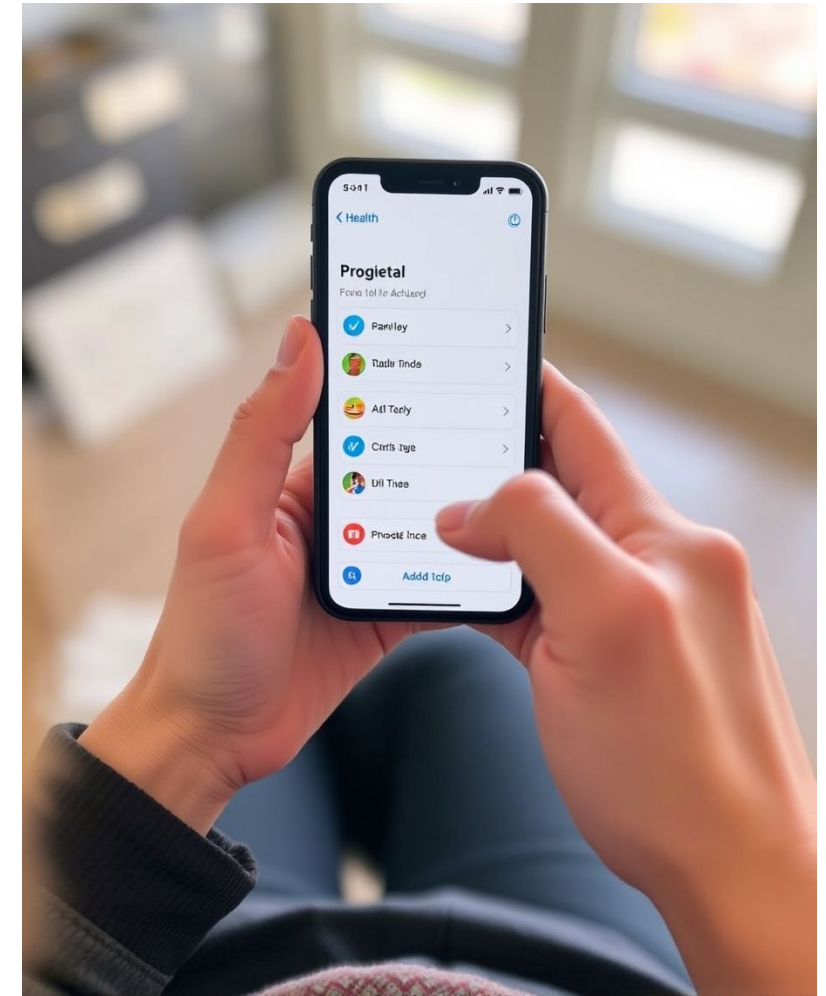


Simplify Task Structures

Concept: Users should not be overwhelmed by complex task sequences. Simplifying the task makes it easier to complete with fewer steps.

Example:

Sarah logs her meal in the health app. Instead of typing in every detail manually, the app suggests previously logged meals with a one-tap option to add them.





Get the Mapping Right (User Mental Model = Conceptual Model = Designed Model)

Concept: The design should match the user's mental model, ensuring they can intuitively understand how to navigate the interface without extensive learning.

Example:

Sarah uses the app to monitor her heart rate. The heart rate is color-coded (green for optimal, red for high), which matches her understanding of healthy heart rates from past experience.

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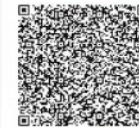
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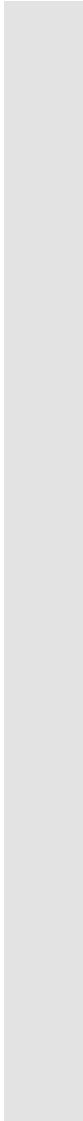
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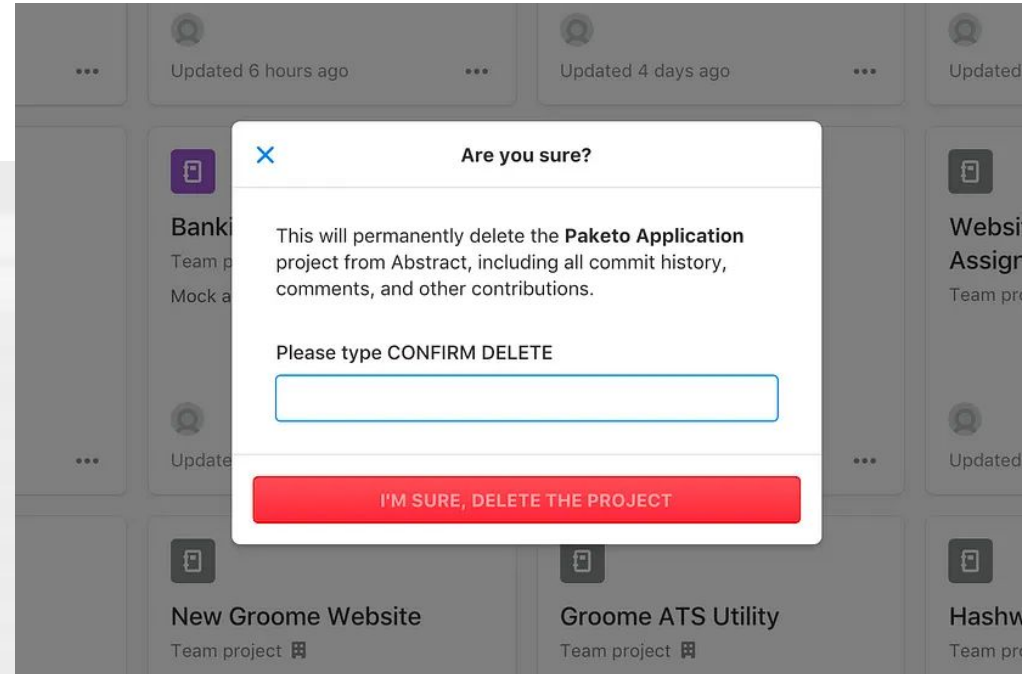
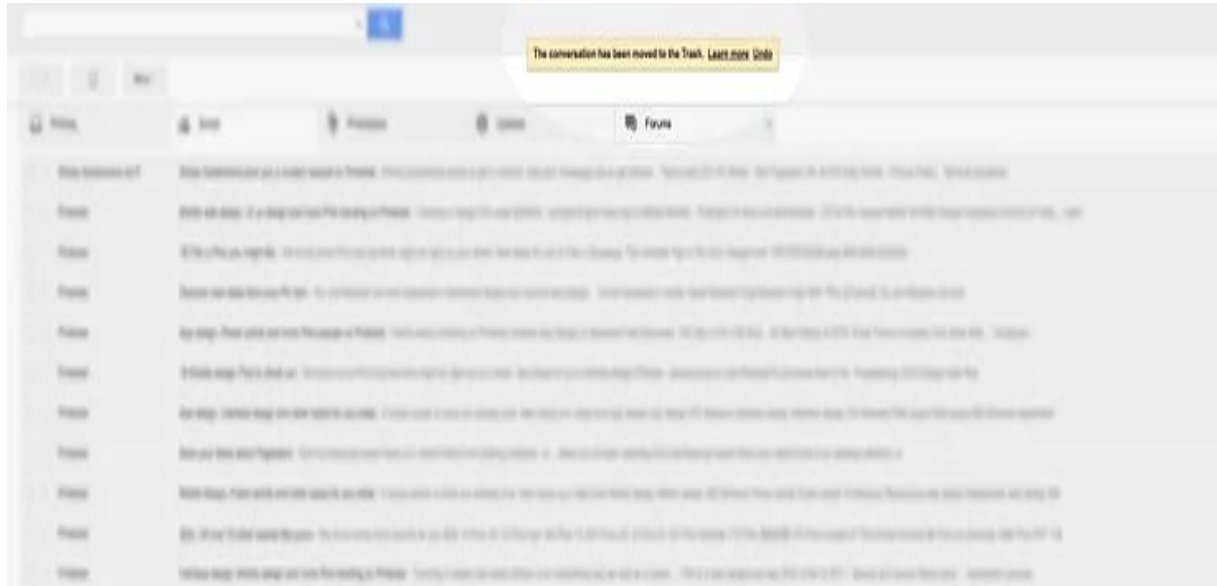
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Messages

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- **Convert Constraints into Advantages (Physical, Cultural, Technological)**
 - **Concept:** Constraints can be turned into design advantages to guide behavior or simplify user tasks.

Design for Error

- Concept: Anticipate and design for user errors by making them easy to detect and correct.
- Example:
- When Sarah accidentally deletes a meal entry, the app provides an undo option immediately on the screen.



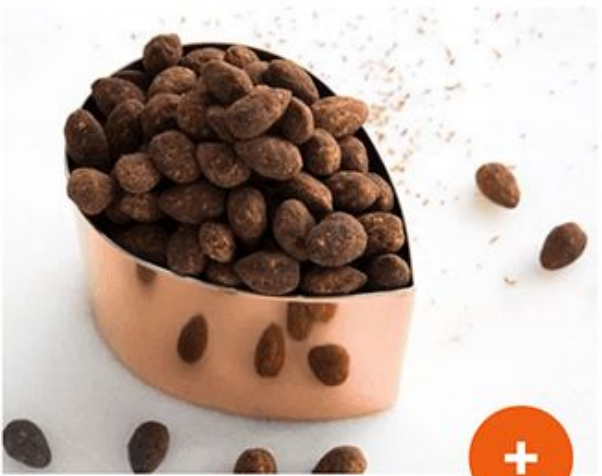
- **When All Else Fails – Standardize**
- **Concept:** Use common conventions and standards to ensure users can easily learn and use the system.
- **Example:**
- Sarah uses the app's navigation menu, which follows the familiar **tab-based design** found in many apps (Home, Steps, Meals, Heart Rate).

- **Make Things Visible**

- Concept: Critical information should be visible to users at all times to reduce the need for memory retrieval.
- Example:
- Sarah's daily step progress is always visible on the app's home screen as a circular progress bar. She doesn't need to navigate through menus to find it.

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